

11 Aperiodic signals, CTFT

Today we'll learn how to represent signals that *don't* repeat.

Continuous time Fourier Series only represent periodic signals, since they integrate over one period.

Can we consider a function over a finite duration, and write it as a sum of $e^{j\omega t}$?

We can copy-paste our signal, and make an envelope so that the front and end agree (don't jump)!

We look at $x(t)$ and tile it to have period over T , (such that they don't overlap).

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

where

$$\omega_0 = \frac{2\pi}{T}$$

$$a_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \tilde{x}(t) e^{-jk\omega_0 t} dt$$

We note that this looks the same as only our original function, since it's just over the length that our function exists (its zero everywhere else)

$$a_k = \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt$$

We define a new thing

$$(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

and say that the thing in our a_k is just $(jk\omega_0)$

Since $\omega_0 = \frac{2\pi}{T}$, $\frac{1}{T} = \frac{\omega_0}{2\pi}$

$$a_k = \frac{1}{T} (jk\omega_0)$$

so we can sub into the synthesis equation

$$\begin{aligned} \tilde{x}(t) &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \\ &= \sum_{k=-\infty}^{\infty} \frac{1}{T} (jk\omega_0) \omega_0 e^{jk\omega_0 t} \end{aligned}$$

Lets take $T \rightarrow \infty$

We now have

$$\tilde{x}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} (jk\omega_0) e^{jk\omega_0 t} \omega_0$$

Lets define another new object

$$x(jk\omega_0) e^{jk\omega_0 t} \quad (k\omega_0, t)$$

$$\tilde{x}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} (k\omega, t) \omega_0$$

As $T \rightarrow \infty$, $\omega = \frac{2\pi}{T}$ small

Result

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega) e^{j\omega t} d\omega$$

We Because $\tilde{x}(t) = x(t)$ as $T \rightarrow \infty$.

This looks "like" the CTFS synthesis. This is the inverse Fourier transform.

$$(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

This is the continuous time Fourier Transform.

For periodic signals, we used $e^{jk\omega_0 t}$, which is countably infinite basis functions - we only have integer k .

For aperiodic signals, The Fourier Transform uses ω everywhere - uncountably infinite basis functions.

Note, our signal has to have finite power, the integrals must converge.

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

1. $H(j\omega)$ is the CTFT of $h(t)$, an aperiodic function.
2. If $x_1(t) = x_2(t)$, then $x_1(j\omega) = x_2(j\omega)$
- 3.